

Program : Diploma in Chemical Engineering	
Course Code : 3079	Course Title: Environmental Engineering Lab I
Semester : 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L: 0 T: 0 P: 3)	Periods per semester: 45

Course Objectives:

- To enhance the analytical ability of students considering health, safety and environment

Course Pre-requisites:

Topic	Course code	Course name	Semester
Air, water, soil, noise pollution, and control measures		Environmental Science	2

Course Outcomes :

On completion of the course student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Evaluate the physical properties of drinking water	12	Applying
CO2	Estimate the chemical properties of drinking water.	21	Applying
CO3	Estimate biological characteristics of drinking water.	6	Applying
CO4	Comprehend air quality monitoring.	6	Applying

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3	3		
CO2				3	3		
CO3				3	3		
CO4				3	3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcome	Description	Duration (Hours)	Cognitive Level
CO1	Evaluate the physical properties of drinking water		
M 1.01	Determination of colour.	1	Applying
M 1.02	Determination of odour.	1	Applying
M1.03	Determination of turbidity.	1	Applying
M1.04	Determination of suspended solids.	3	Applying
M1.05	Determination of dissolved solids.	3	Applying
M1.06	Open ended experiments	3	Applying
CO2	Estimate the chemical properties of drinking water.		
M2.01	Determination of pH.	1	Applying
M2.02	Determination of conductivity.	2	Applying
M2.03	Determination of total hardness of water by EDTA method.	3	Applying
M2.04	Determination of total alkalinity.	1	Applying
M2.05	Determination of acidity of water	2	Applying
M2.06	Determination of sulphate.	3	Applying
M2.07	Determination of residual chlorine.	3	Applying
M2.08	Determination of iron content in water.	3	Applying
M2.09	Open ended experiments	3	Applying
	Series Test – I		

CO3	Estimate biological characteristics of drinking water.		
M3.01	Estimate the presence of E-coli in water.	3	Applying
M3.02	Open ended experiments	3	Applying
CO4	Comprehend air quality monitoring.		
M4.01	Estimate the amount of particulate matter in ambient air.	3	Applying
M4.02	Estimate the amount of CO, CO ₂ and O ₂ using Orsat apparatus.	3	Applying
	Series Test – II		

Text / Reference

T/R	Book Title/Author
T1	Wastewater engineering: treatment and reuse by Metcalf & Eddy
R1	Principles and Practices of Air Pollution Control and Analysis by J R Mudakavi
R2	Standard Methods for the Examination of Water and Wastewater, 22nd Edition
R3	Environmental engineering volume 1; S.K Garg

Online Resources

Sl.No	Website Link
1	https://www.vlab.co.in/broad-area-civil-engineering